

In the eastern part of Prince Edward Island, drilling has revealed at least 350 feet of strata that contain *Nuskoisporites*, *Vittatina*, *Lueckisporites*, and *Platysaccus*, and is considered to be Permian in age. The relationship of these sediments to the Stephanian sediments of western Prince Edward Island is presently unknown.

A generic range chart, based principally on previous spore studies in the Maritime Provinces, is included. Certain significant genera, e. g., *Densosporites*, and *Vestispora*, are shown to have vertical distribution which is different from that observed in the United States and Europe.

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Suggested notations to indicate identification status of fossil pollen.

Names of living taxa applied to fossil pollen and spores commonly become separated from whatever qualifying descriptions and illustrations were used in the text of the original publication, so that the status of certainty of identification fails to accompany each name in every place it appears. Practices of North American pollen analysts differ widely with respect to reporting the status of pollen and spore identifications. We suggest that consideration be given to use of a graded series of conventions to indicate degrees of certainty in identification of those pollen grains and spores that can be appropriately referred to living taxa. The following list of conventions is arranged in order of decreasing certainty of identification :

*Taxon* — (with no qualifying notation about identification status) identification certain, i. e., the pollen grain or spore of no other taxon (at least in this flora) could be confused with this one.

*Taxon id.* — (abbreviation for *idem*, *eadem*, *idem*, identical) identification certain ; used in instances in which the fact of positive identification needs to be indicated.

*Taxon comp.* — (abbreviation for *comparatus*, *-a*, *-um*, comparative) compares favorably with material from living representatives of taxon to which entity is provisionally assigned, but uncertainty exists about conclusiveness of identification.

*Taxon sim.* — (abbreviation for *similis*, *-e*, resembling, similar) strong similarity to indicated taxon, which is, or is suspected of being, representative of several indistinguishable taxa of comparable rank with respect to pollen or spore morphology.

Undetermined — definitive morphological characters present, but taxonomic relationships unknown. Within this category several recognizable groups may be distinguished and named according to a purely morphologic scheme, accompanied by descriptions and illustrations.

Unknown — entities for which morphological characters are insufficient for determining whether they are pollen grains, spores, or other bodies.

The symbol -/- (hyphen, slant, hyphen), denoting damaged specimens, may be added to any of the expressions for status of identification.

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Calculation of pollen and spores density in sediments by addition of exotic pollen in known quantities.

Absolute population density of fossil pollen and spores in sediments may be calculated in preparations to which a known volume of a standardized suspension of exotic pollen has been added before chemical processing. One or several kinds of exotic pollen grains or spores (i. e., kinds which would not normally occur in the sediments under study) are suspended in glycerine, which is moderately viscous and slow to evaporate. Population density of each exotic grain in the suspension is determined by counting grains per unit volume in a microscope slide counting cell. A measured volume of the suspension is added to a known volume (measured as centrifuged sediment in a graduated tapered tube) of the sediment under study. The mixture of sediment and added suspension is washed twice with distilled water in a centrifuge tube, and then subjected to standard procedures for fossil pollen preparations. While the fossil pollen and spores in the final preparation slides are being counted, the added exotic grains are also counted and tabulated. As the number of exotic

grains added to the original sample can be calculated from their density in the suspension and the volume of the suspension, the number counted in the final preparation permits calculation of the proportionate volume of the original sediment sample that has been inspected. The total number of fossil pollen and spores counted constitutes the population density of that proportionate volume of the original sediment, and this value may be expressed in terms of total pollen and spore density per 1 cc. or other standard unit volume.

In theory, if the pollen in the additive suspension were of two widely different size ranges, 80 microns and 10 microns for example, any change in the proportion of the two sizes types in the final preparations would indicate fractionation or differential loss of certain particle sizes during preparation procedures. Experience with these techniques has shown that the limits of error in sampling and counting need to be established at each step in the procedures. In some instances the compounding of relatively wide limits of error in the several steps seriously reduces the reliability of final calculations ; but some of these experiences are instructive with respect to common practices with pollen statistics.

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Pollen studies of vegetational history in the Chuska Mountains, New Mexico.

The Chuska Mountains in northwesternmost New Mexico have a flat crest 2700 m above sea level covered with *Pinus ponderosa* with groves of *Quercus gambeli* and *Populus tremuloides*. Canyons locally bear *Picea pungens*, *Abies lasiocarpa*, *Pseudotsuga taxifolia*, and *Populus tremuloides*. Below about 2300 m is a woodland of *Pinus edulis* and *Juniperus*, with a shrub cover of *Artemisia tridentata*. Below about 2000 m east of the mountains is a sparse cover of Compositae, Chenopodiaceae, and other low semidesert shrubs.

Pollen analyses of surface samples from the crest, flank, and base of the mountains show a reasonable correspondence between pollen precipitation and surrounding vegetation. Pine is of course the dominant pollen type in all samples, but high percentages of Chenopod and locally Juniper or *Artemisia* pollen are diagnostic of basin sites, whereas crestal sites have higher percentages of *Picea*, *Abies*, *Pseudotsuga*, and locally *Quercus*. The best index for altitude is the ratio of *Picea* + *Abies* + *Pseudotsuga* to *Chenopodiaceae* + *Ephedra*.

Hundreds of small undrained depressions dot the flat top of the range, which is capped by siliceous sandstone. The sediment in the depressions consist of a few decimeters of black clay underlain by several meters of gray clay or sandy clay. The top black clay is dominated by pollen of *Pinus*, with some *Quercus* and Chenopods, but very little *Picea* and little *Artemisia*, representing vegetation much like the present. A C-14 date for the lower part of the black layer at one lake reads about 3600 BP (L-515A), and the black layer is considered to represent much of the Holocene.

The underlying gray sediment, which reaches a thickness of 11 m in Deadman Lake, has a strikingly different pollen content. Although *Pinus* maintains values of 20-40 % of total pollen, *Picea* is 10-20 % and *Artemisia* 30-60 %. *Quercus* is much lower than in the top sediment, and chenopods are somewhat lower. Today the Spruce forest generally occurs at higher elevations than *Pinus ponderosa* in the western mountains. The gray sediment is therefore believed to date from the Pleistocene when the tree zones were depressed a few thousand feet below the present position. The high values of the *Artemisia* pollen may therefore not represent primarily *A. tridentata* (sagebrush), which now occurs many thousand feet below the spruce zone, but rather subalpine species. The mountain crest may have been far enough above the pine zone that there were few other large pollen producers in the region. The Pleistocene correlation of the gray sediment is confirmed by a C-14 date of > 28,000 BP (A-213) at Deadman Lake at a depth of 9.30 m. Additional pollen-analytical and other paleontological studies of the lakes are being made to refine the Pleistocene and Holocene climatic succession.

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Pollen analytical arguments for plant migrations in south Europe.